

A Partial Measure-Extraction Closure Theorem for Problem 7.3 of arXiv:2207.00149

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Source Problem

The source paper is Damian Glodkowski, *A Banach space $C(K)$ reading the dimension of K* , arXiv:2207.00149. In Section 7, after using the metrizable measure-extraction lemma in the construction, the paper asks:

Problem 7.3. Describe the class of compact Hausdorff spaces K such that for every non-zero Radon measure μ on K there is a zero-dimensional compact subset $L \subseteq K$ such that $\mu(L) \neq 0$.

intermediate spaces from our construction are metrizable. At that point we used the fact that for every non-zero Radon measure on metrizable compact space there is a zero-dimensional G_δ compact subset of non-zero measure (Theorem 3.16). In the light of this theorem the following problem seems to be interesting.

Problem 7.3. *Describe the class of compact Hausdorff spaces K such that for every non-zero Radon measure μ on K there is a zero-dimensional compact subset $L \subseteq K$ such that $\mu(L) \neq 0$.*

Assume that K is such that $C(K)$ has few operators. Then by [36, Proposition 4.8] there is a space L such that $C(K) \sim C(L)$, but $C(L)$ does not have few operators. However, by Theorem 4.19 the topology of L is very close to K , at least if we assume that K is separable and connected.

Result

This packet proves a closure theorem for a strong form of the property in Problem 7.3. It is not a full solution of the problem, and it does not remove the set-theoretic hypothesis from the main theorem of the source paper. It does show that the measure-extraction property is compatible with the exact extension operation used in the paper when the exceptional set is already zero-dimensional.

Definition 1 (Strong zero-dimensional measure extraction). *A compact Hausdorff space K has property SZM if for every finite positive Radon measure μ on K and every $\eta > 0$, there is a compact zero-dimensional set $Z \subseteq K$ such that*

$$\mu(K \setminus Z) < \eta.$$

Theorem 1. *The class of compact Hausdorff spaces with property SZM has the following closure properties.*

1. It is inherited by compact subspaces.
2. It contains all compact metrizable spaces and all zero-dimensional compact spaces.
3. It is closed under finite products.
4. Let L be an extension of K by a pairwise disjoint sequence $(f_n) \subseteq C(K, [0, 1])$, with natural projection $\pi : L \rightarrow K$. Let $D = D((f_n))$ be the open set on which $\sum_n f_n$ is continuous, so that π is a homeomorphism from $\pi^{-1}(D)$ onto D . If K has SZM and $E = K \setminus D$ is zero-dimensional, then L has SZM.

Corollary 1. *Every finite tower of the source-paper extension steps, starting from a compact metrizable space and using only steps for which $K_\alpha \setminus D((f_n^\alpha))$ is zero-dimensional, has property SZM.*

Idea of the Proof

The metrizable proof in the source paper removes the boundaries of a countable family of neighbourhoods and leaves a positive-measure zero-dimensional remainder. The closure theorem isolates the part of that argument that survives without metrizability: once a compactum has enough zero-dimensional compact sets to approximate all Radon measures, products preserve this by projecting the measure one coordinate at a time. For an extension $L \subseteq K \times [0, 1]$, the measure splits into the graph part over D , which is homeomorphic to a subspace of K , and the exceptional part over $E = K \setminus D$, which sits inside $E \times [0, 1]$. The first part is handled by heredity; the second by the product theorem, because E is zero-dimensional and $[0, 1]$ is metrizable.

Proof

Lemma 1 (Heredity). *If K has SZM and $F \subseteq K$ is compact, then F has SZM.*

Proof. Let ν be a finite positive Radon measure on F . Extend it to a Radon measure $\tilde{\nu}$ on K by the positive functional $g \mapsto \int_F g|_F d\nu$ on $C(K)$. Given $\eta > 0$, choose compact zero-dimensional $Z \subseteq K$ with $\tilde{\nu}(K \setminus Z) < \eta$. Then $Z \cap F$ is compact and zero-dimensional, and

$$\nu(F \setminus (Z \cap F)) = \tilde{\nu}(K \setminus Z) < \eta.$$

□

Lemma 2 (Basic examples). *Every compact metrizable space and every zero-dimensional compact Hausdorff space has SZM.*

Proof. For zero-dimensional K , take $Z = K$. Now let K be compact metrizable and let μ be a finite positive Radon measure. Fix a compatible metric and a countable dense set $\{d_n : n \in \mathbb{N}\}$. For each n , choose a sequence of distinct radii $r_i^n > 0$ such that every tail $\{r_i^n : i \geq k\}$ is dense in $(0, \text{diam } K)$, and put $U_i^n = B(d_n, r_i^n)$. For fixed n , the boundaries ∂U_i^n are pairwise disjoint, so choose k_n with

$$\sum_{i \geq k_n} \mu(\partial U_i^n) < \eta 2^{-n-2}.$$

Let

$$Y = K \setminus \bigcup_{n=0}^{\infty} \bigcup_{i \geq k_n} \partial U_i^n.$$

Then $\mu(K \setminus Y) < \eta/2$. The sets $U_i^n \cap Y$, for $i \geq k_n$, form a base of Y consisting of clopen subsets, so Y is zero-dimensional. By inner regularity, choose compact $Z \subseteq Y$ with $\mu(Y \setminus Z) < \eta/2$. Then Z is compact zero-dimensional and $\mu(K \setminus Z) < \eta$. $\square$

Lemma 3 (Finite products). *If K and M have SZM, then $K \times M$ has SZM.*

Proof. Let μ be a finite positive Radon measure on $K \times M$, and let $\eta > 0$. Let $\nu = (p_K)_*\mu$. Choose compact zero-dimensional $A \subseteq K$ such that $\nu(K \setminus A) < \eta/2$. Restrict μ to $A \times M$ and project to M , obtaining a finite positive Radon measure ρ . Choose compact zero-dimensional $B \subseteq M$ with $\rho(M \setminus B) < \eta/2$. Then $A \times B$ is compact zero-dimensional and

$$\mu((K \times M) \setminus (A \times B)) \leq \mu((K \setminus A) \times M) + \mu(A \times (M \setminus B)) < \eta.$$

$\square$

Proof of the theorem. The first three assertions are the preceding lemmas and induction for finite products. It remains to prove the extension assertion.

Let μ be a finite positive Radon measure on L , and fix $\eta > 0$. Put $L_D = \pi^{-1}(D)$ and $L_E = \pi^{-1}(E)$. The set L_E is compact and is a compact subspace of $E \times [0, 1]$. Since E is zero-dimensional and $[0, 1]$ is compact metrizable, $E \times [0, 1]$ has SZM by the product lemma, hence L_E has SZM by heredity. Choose compact zero-dimensional $Z_E \subseteq L_E$ such that

$$\mu(L_E \setminus Z_E) < \eta/2.$$

For the graph part, use inner regularity to choose compact $C_D \subseteq L_D$ such that

$$\mu(L_D \setminus C_D) < \eta/4.$$

Because $\pi|_{L_D}$ is a homeomorphism onto D , the compact set $H = \pi[C_D]$ is a compact subspace of K , and C_D is homeomorphic to H . By heredity, H has SZM. Apply this property to the push-forward of $\mu|_{C_D}$ under π and choose compact zero-dimensional $A \subseteq H$ so that

$$\mu(C_D \setminus \pi^{-1}(A)) < \eta/4.$$

Then $Z_D = \pi^{-1}(A) \cap C_D$ is compact zero-dimensional, and

$$\mu(L_D \setminus Z_D) < \eta/2.$$

The compact sets $Z_D \subseteq L_D$ and $Z_E \subseteq L_E$ are disjoint. In a Hausdorff compact space two disjoint compact sets are separated by open sets, so their union is again zero-dimensional. Thus

$$Z = Z_D \cup Z_E$$

is compact zero-dimensional and $\mu(L \setminus Z) < \eta$. This proves that L has SZM. $\square$

Signed and Complex Measures

The theorem is stated for finite positive Radon measures. The non-zero signed or complex variants in Problem 7.3 follow from the positive version in the standard way: take a non-zero real part after rotation if necessary, choose a Hahn-positive compact set of positive variation, and apply the hereditary positive theorem inside that compact set. The resulting zero-dimensional compact subset has non-zero value for the original measure after the chosen rotation is undone.

Limitations

This partial result does not yet solve the main ZFC problem from the source paper. Proposition 5.9 in the source paper uses not only a positive-measure zero-dimensional compact set but a countable decreasing neighbourhood sequence around it, available automatically in compact metrizable spaces. The closure theorem above gives compact zero-dimensional subsets but does not guarantee that they are G_δ or have countable character in the ambient compactum. Long inverse-limit stages are also not covered by the finite closure argument.

Novelty and Literature Check

The run indexes had no prior packet or attempt for arXiv:2207.00149 or Problem 7.3. Searches were run for the exact title, the exact problem phrase, and variants including “zero-dimensional compact subset non-zero Radon measure”, “Radon measure compact Hausdorff zero-dimensional subset”, and Maharam/uniform-regularity formulations. No separate paper explicitly answering Problem 7.3 was found. Related results on uniformly regular measures and simple extensions were noted, but they do not remove the G_δ /countable-character obstacle identified above.

References

1. Damian Glodkowski, *A Banach space $C(K)$ reading the dimension of K* , arXiv:2207.00149, 2022.
2. Ryszard Engelking, *Theory of Dimensions, Finite and Infinite*, Heldermann Verlag, 1995.
3. M. Dzamonja and G. Plebanek, *On Efimov spaces and Radon measures*, Topology Appl. 154 (2007), 2063–2072.

Human Review Recommendation

Review as a likely valid partial theorem. The most important follow-up is to test whether SZM can be strengthened to return a zero-dimensional compact G_δ subset under the extension and limit operations relevant to the ZFC few-operators construction.